Finding the Distance Between Points

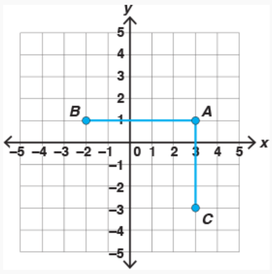
You can use a coordinate grid to find the distance between two points. The *x*-coordinate will help you find horizontal distance. The *y*-coordinate will help you find vertical distance.

Example 1

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What is the distance from *A* to *B*?

Point *A* is 3 units from the *y*-axis. Point *B* is 2 units from the *y*-axis. The total distance is 3 + 2 = **5.**



distance from A to B =

$|AB| = 3-(-2)$

$= 3+2$

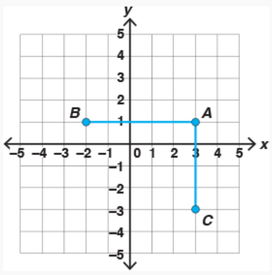
$= 5$

Example 2

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What is the distance from *A* to *C*?

Point *A* is 1 unit from the *x*-axis. Point *C* is 3 units from the *x*-axis. The total distance is 1 + 3 = **4.**



$|AC| = 1-(-3)$

$= 1+3$

$= 4$

You can also subtract the coordinates and take the absolute value.

In the first example, subtract the *x*-coordinates. $|-2-3| = |-5| = 5$

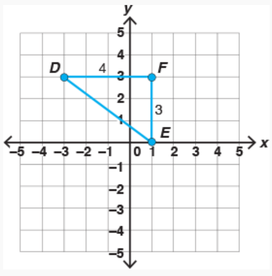
In the second example, subtract the y-coordinates. $|1--3| = |1+3| = |4| = 4$

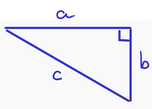
Once you find the horizontal and vertical distances, you can use the Pythagorean theorem to find any distance on a coordinate plane.

Example 3

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What is the distance from *D* to *E*?





$c^2 = a^2 + b^2$

$c = \sqrt{a^2 + b^2}$

$|DE| = \sqrt{(1-(-3))^2 + (0-3)^2}$

$= \sqrt{4^2 + (-3)^2}$

$= \sqrt{16+9}$

$= \sqrt{25}$

$= 5$

Step 1

By drawing horizontal and vertical lines, you can create a right triangle where *DE* is the hypotenuse. Find the lengths of *DF* and *EF*.

$DF = |-3-1| = |-4| = 4$

$EF = |0-3| = |-3| = 3$

Step 2

Use the Pythagorean theorem.

$a^2 + b^2 = c^2$

$3^2 + 4^2 = c^2$

$\sqrt{9+16} = c$

$\sqrt{25} = c$

$5 = c$

The distance from *D* to *E* is **5 units.**

There is a formula for finding the distance between two points that combines the process of subtracting the coordinates and the Pythagorean theorem. You don't have to memorize it because you already know the Pythagorean theorem. Take a moment to investigate the formula.

Distance $(d) = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$, where (x_1, y_1) and (x_2, y_2) are points

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